Northern European Forests' carbon balance seen by EC-towers: a synthesis

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ICOS Finland annual meeting 27.04.2026

Ränskälänkorpi clear-cut tower on fertile drained peatland, Southern Finland. Operated by Luke 2021→

Why?

Forests:

- ✓ face multiple demands – providing carbon storage, climate regulation, income and well-being

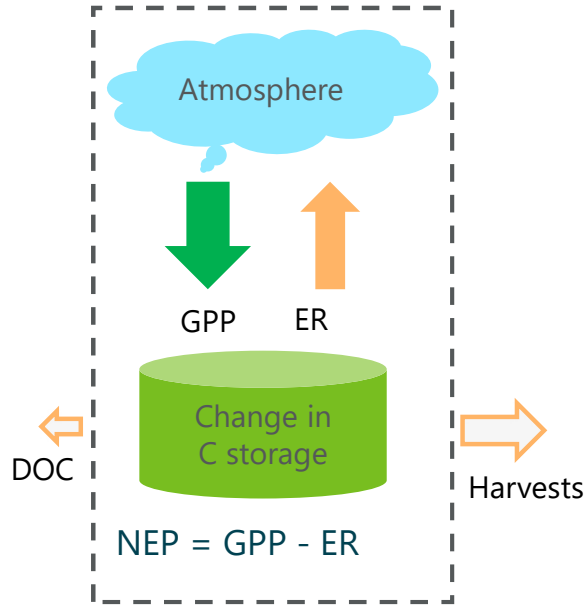
Northern Europe:

- ✓ forests and forestry major land use & crucial for national climate targets
- ✓ Active flux community
- ✓ Reasonably uniform – species, soils, climate, management

Synthesis:

- ✓ So far very few (Lindroth et al. 2008, 2020)
- ✓ So far only a few sites included

Ecosystem (on-site) carbon balance



Net ecosystem productivity NEP = - NEE from eddy-covariance

Ecosystem respiration ER = f(T), based on night-time NEP

Gross primary productivity GPP = NEP + ER

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What is out there?

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Forest C balance by eddy-covariance?

Northern European forests	ICOS warm winter dataset FluxNet 2015 Literature review
Minimum 1 year of NEP data	Site attributes → species, (LAI), soil type, site type, management history Habitat, Köppen climate ERA5 Land → climate, gs length
Published & citable	Minimum a MSc thesis or preprint

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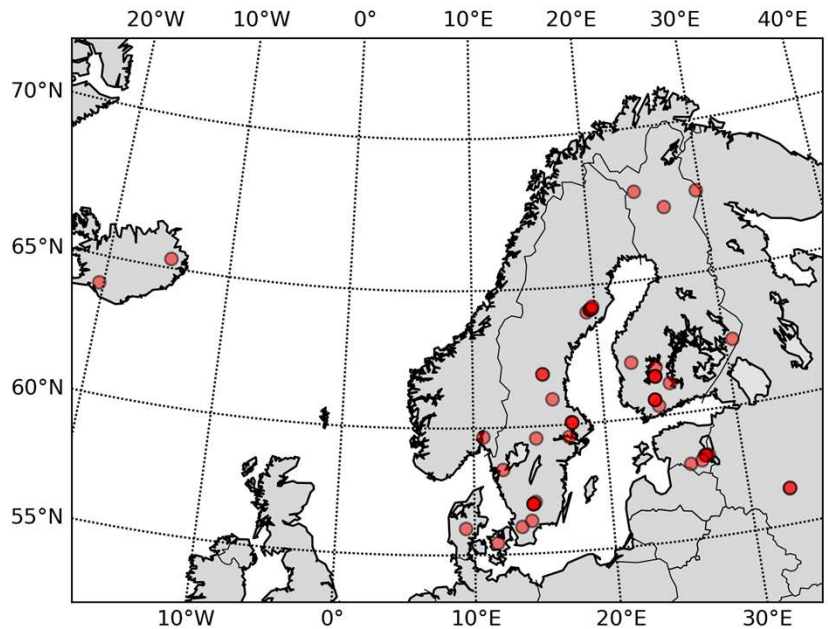
NEP:
50 sites
320 site-years
1 to 26 yr/site

GPP/ER:
44 sites
252 site-years

1997 – 2022

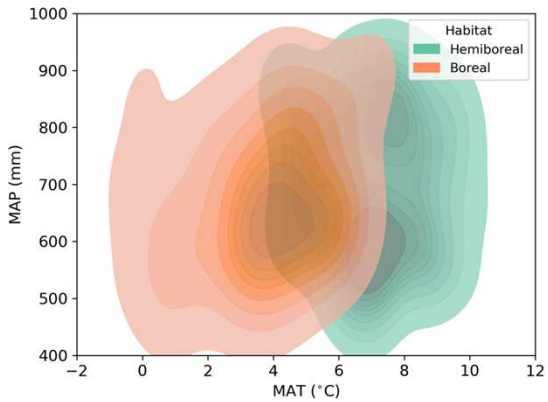


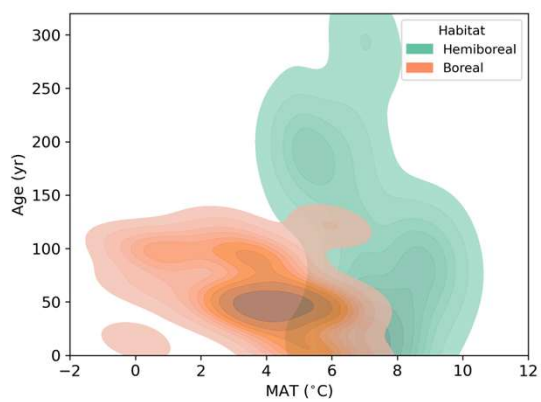
Sweden 22
Finland 15
Estonia 7
Denmark 2
Iceland 2
Russia 2
Norway 1

11 peatland forests
3 plantations
2 post-fire sites

11 recent clear-cuts
3 thinned sites
2 N-fertilization

10 ICOS warm-winter
1 Fluxnet 2015
39 Literature review





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How carbon balance vary with stand age?

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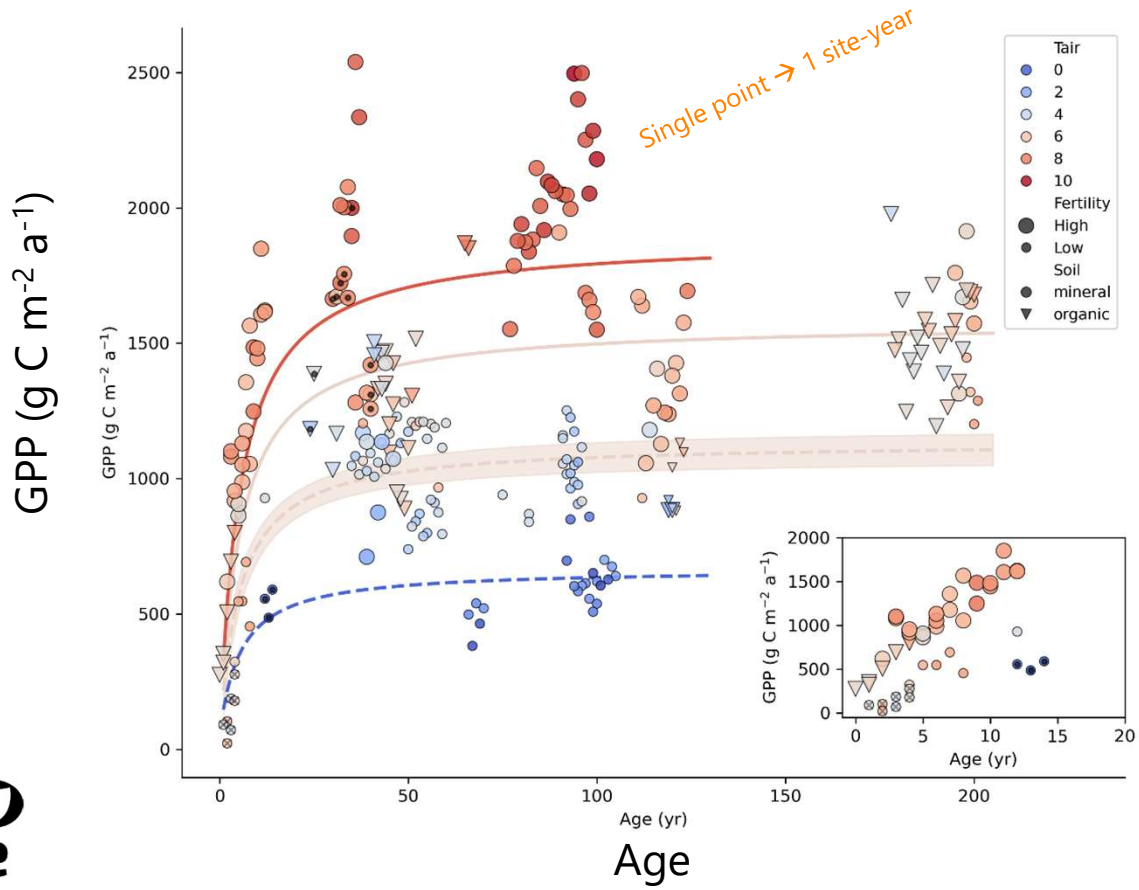
Gross-primary productivity



$$GPP = \epsilon + b \times \frac{Age}{Age + c}$$

Age only: 17% of variance

Age, T_a and site attributes: 78%

$b \propto T_a$ +

$+ VPD_{gs}$ −

$+ fertility(low)$ −

color = MAT
size = fertility
symbol = soil type
dot = plantation
cross-hatch = post-fire

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Ecosystem respiration

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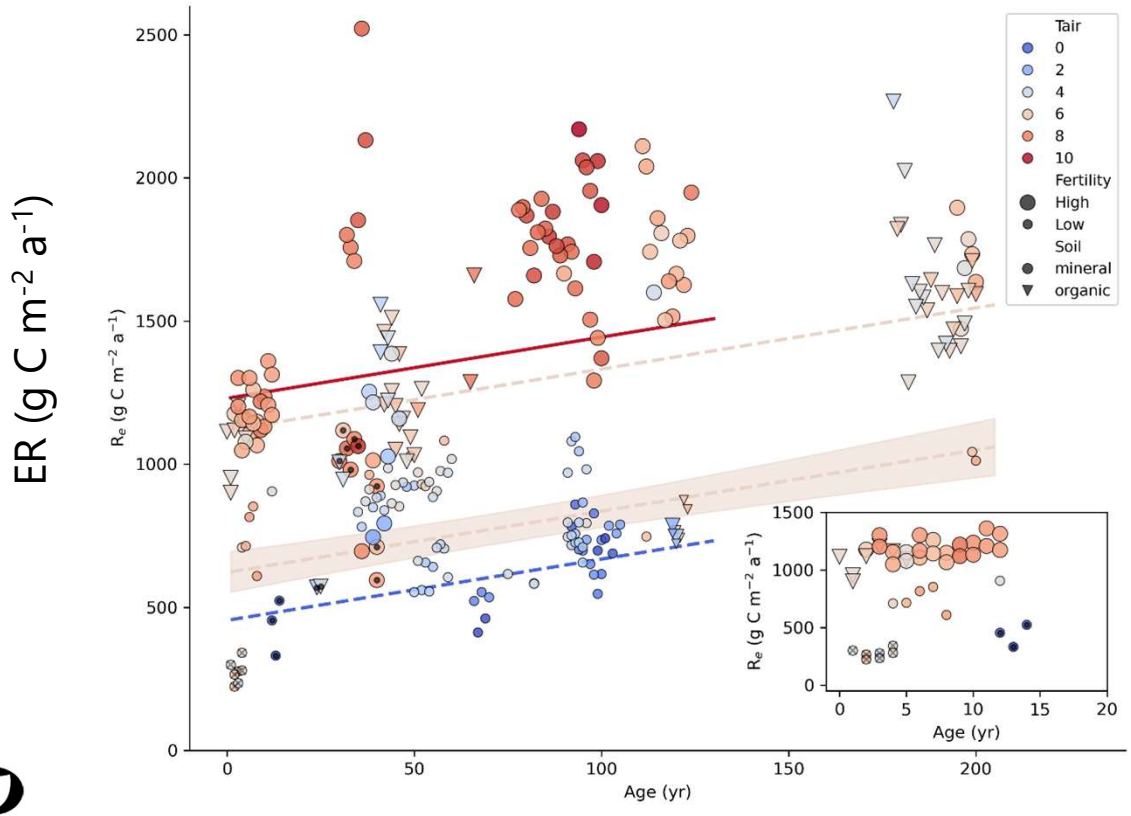

Age

$$R_e = \epsilon + a + d \times Age$$

Age only: 13%

With T_a and site attributes: 71%

$$a \propto T_a$$ ✚

$$+ plantation\,(yes)$$ ▬

$$+ fertility\,(low)$$ ▬

$$+ habitat\,(hemiboreal)$$ ✚

color = MAT
size = fertility
symbol = soil type
dot = plantation
cross-hatch = post-fire

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NEP – on-site carbon balance

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$$NEP = \epsilon + a + b \times \frac{Age}{Age + c} - d \times Age$$

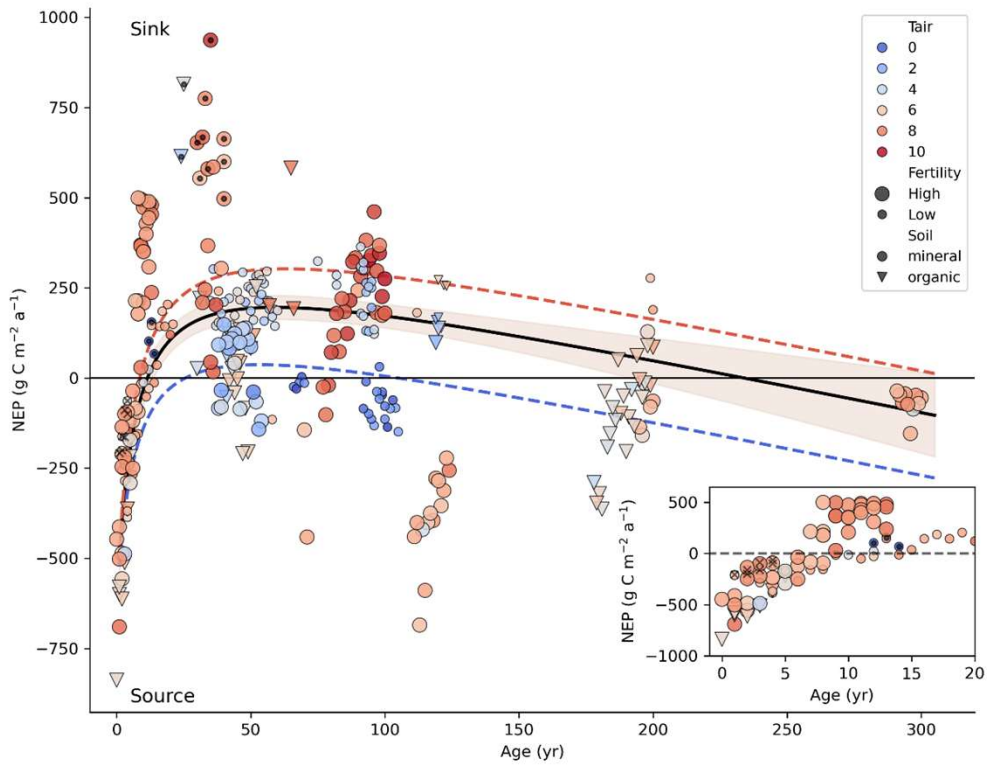
Age only: 18 % of variance

Age, T_a and plantation: 39 %

$$a \propto T_a$$

$$+ plantation\,(yes)$$

NEP (g C m^{-2} a^{-1})



Age

color = MAT
size = fertility
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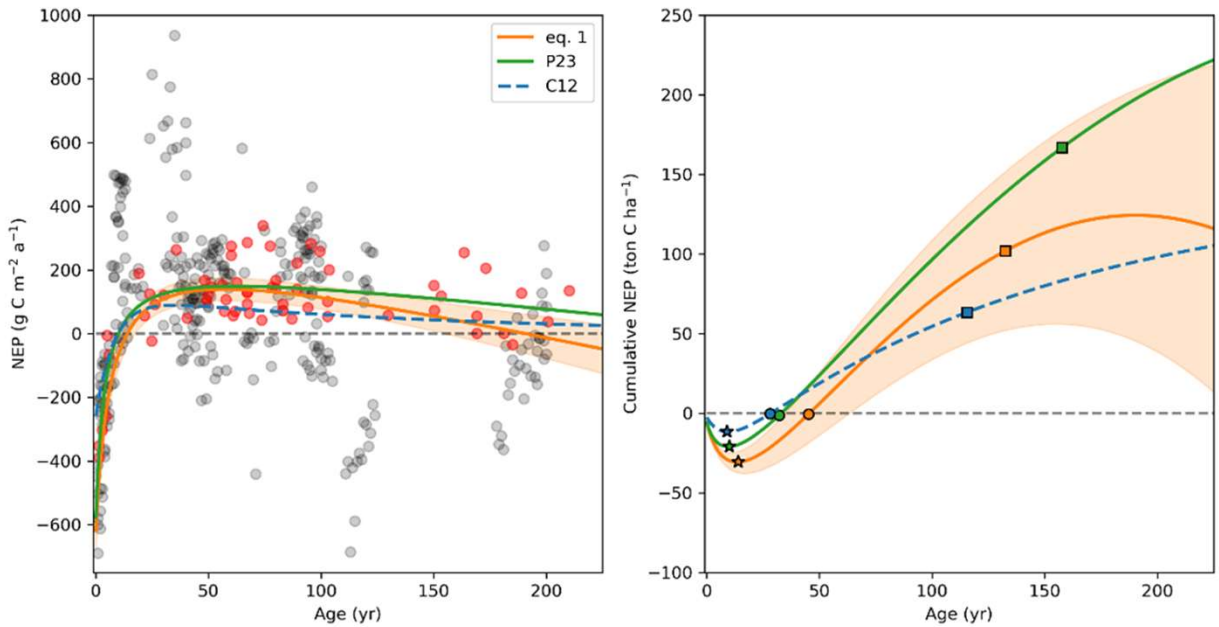
Time-integrated NEP

- Krycklan local (P23, red dots; Peichl et al. 2023)
- Canadian boreal forests (C12, Coursolle et al. 2013)

- NEP-Age dynamics are “similar”
- .. but high between-site / within landscape variability of NEP makes generalizations useless for any policy or management decisions



Stars = source-sink transition (carbon compensation point)
Circles = initial emissions offset (recovery period)
Rectangles = optimal rotation length

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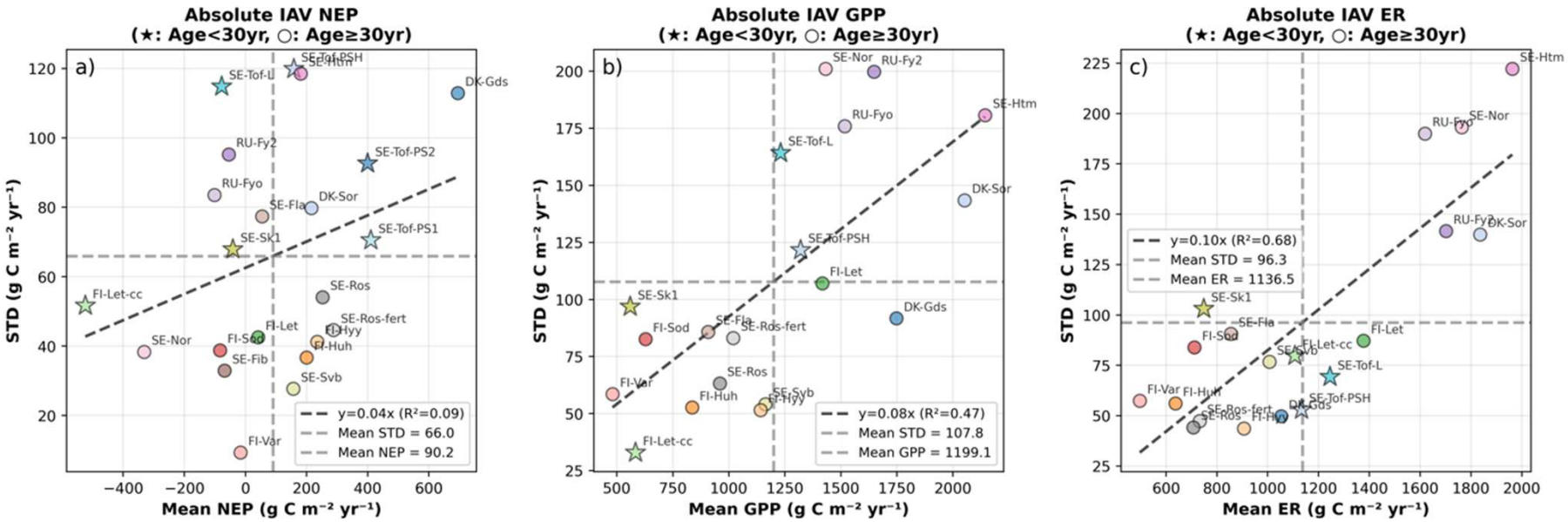
Inter-annual variability



Approach:

- Sites with >5yr data; linear detrending to remove long-term effects
- IAV estimated as std of detrended timeseries

IAV:

NEP± 66 (± 75% of the mean!)
GPP: ± 30 to 200 (±8%)
ER: ± 40 to 225 (±10%)

IAV of GPP and ER is proportional to the mean

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How forest management practices affect C balance?

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Rapid NEP recovery after clear-cutting

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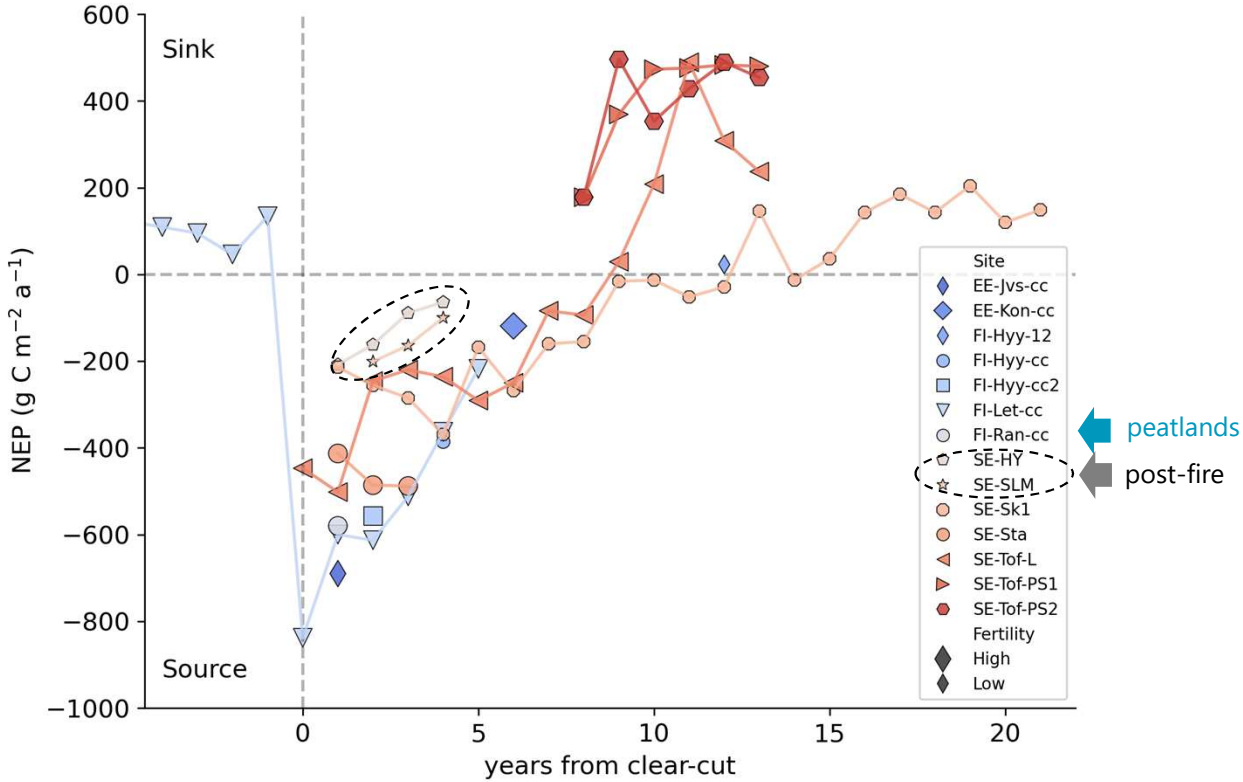
- ✓ From source to sink <10 year post-harvest
- ✓ Initial emissions depend on site fertility and amount & fate of cutting residues
- ✓ Significant GPP immediately after clear-cutting
- ✓ Fast GPP recovery drives NEP dynamics
  - ✓ Deciduous seedlings, grass, herbs, shrubs
- ✓ ER = Ra + Rh changes conservative during first two decades after harvest (compensation)



Kolari et al. (2004); Ahmed (2019); Korkiakoski et al. (2023); Tikkasalo et al. (2024); Kelly et al. (2024); Grelle et al. (2023, 2012); Uri et al. (2019); Rebane et al. (2024)

Thinning response is short-lived

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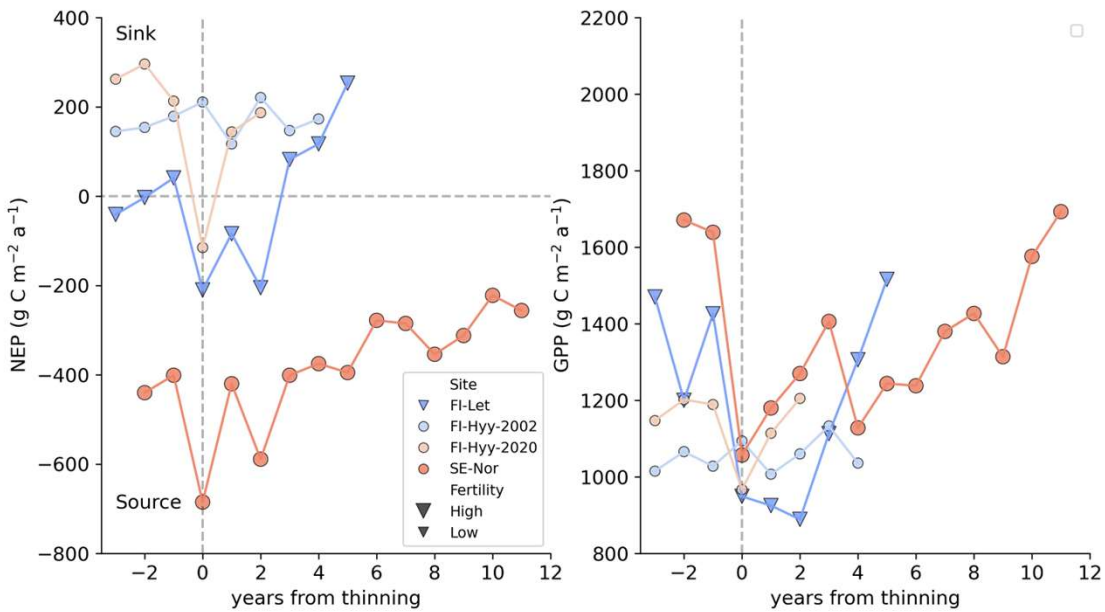

- ✓ Short-term drop in NEP is due to GPP response
- ✓ GPP reduction (%) always smaller than removed LAI (%)
  - ✓ Light (water, nutrient) availability
- ✓ Rapid GPP and NEP recovery
- ✓ Fertile, dense stands (FI-Let & SE-Nor):
  - ✓ ER decreases after thinning
  - ✓ Thinning improves on-site carbon sink

Site	Removed LAI (%)	Removed BA (m^2ha^{-1})	Site details
FI-Let	**50**	**25**	Mixed species, fertile drained peatland (Korkiakoski et al. 2023)
FI-Hyy 2002	**25**	**6**	Mixed conif., 1st thinning, infertile mineral soil (Vesala et al. 2005)
FI-Hyy 2020	**40**	**12**	2nd thinning (Aslan et al. 2024)
SE-Nor	**50**	**10**	fertile old spruce, mineral soil, high organic content (Lindroth et al. 2018)

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N fertilization improves NEP

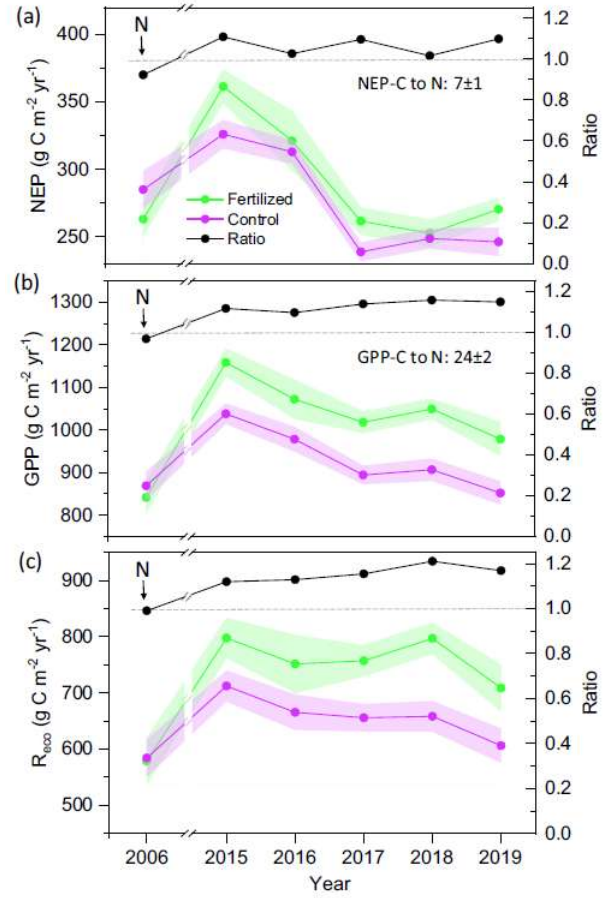
Rosinedal, N. Sweden

✓ Scots pine on low-fertility mineral soil

✓ N-fertilization 100 kg N ha^{-1} a^{-1} (2006-2011), 50-60 kg N ha^{-1} a^{-1} (2012-2019)

✓ Long-term N-fertilization increases both GPP and ER.

✓ Small(er) positive effect on NEP



Zhao et al. (2022) Agric. For. Met.
https://doi.org/10.1016/j.agrformet.2022.109112

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Message – forest management

- ✓ Forestry operations act primarily on GPP
  - ✓ Rapid disturbance recovery
  - ✓ We must re-think the role of pioneering vegetation on C sink; currently poorly quantified and often ignored in models
- ✓ Based on EC-data, peatland forests NEP and its dynamics not different from mineral soil forests
- ✓ Impact to on-site C balance (NEP) only part of the story – the fate and lifetime of harvested C storage
- ✓ Three decades of EC measurements have contributed little for improving forest management

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Holistic infrastructure for knowledge on land-use sector carbon sinks and emissions tracking

Implications?

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Naïve NEP-upscaling to Finland

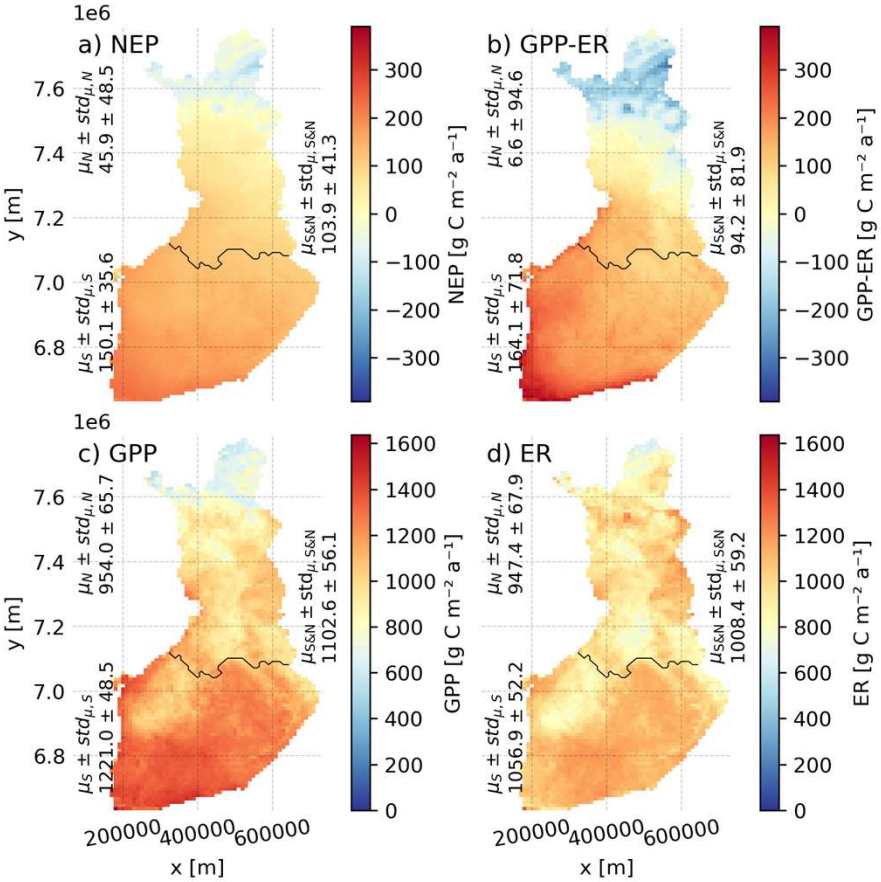
Best models (NEP, GPP – ER) fitted to NE forest flux data used

- Age, site fertility (← latest multi-NFI, 16x16m)
- Annual temperature, growing season VPD (← gridded weather data)

Average NEP over Finnish forest land (g C m^{-2} a^{-1}):

NEP (mean +/- std)	GPP-ER	NEP* (GHG-I)
104 ± 41	94 ± 82	126 ± ?

- NEP = GPP – Ra – Rh = NPP – Rh
- NEP* = (growth + litter prod.) – (litter prod – soil C balance) = growth + soil C balance.
- **NFI13 & GHG-I:** growth = 159, mineral soil balance +3, organic soil balance -34 g C m^{-2} a^{-1}



… but: mean IAV in NE forests dataset was ~66 g C m^{-2} a^{-1}

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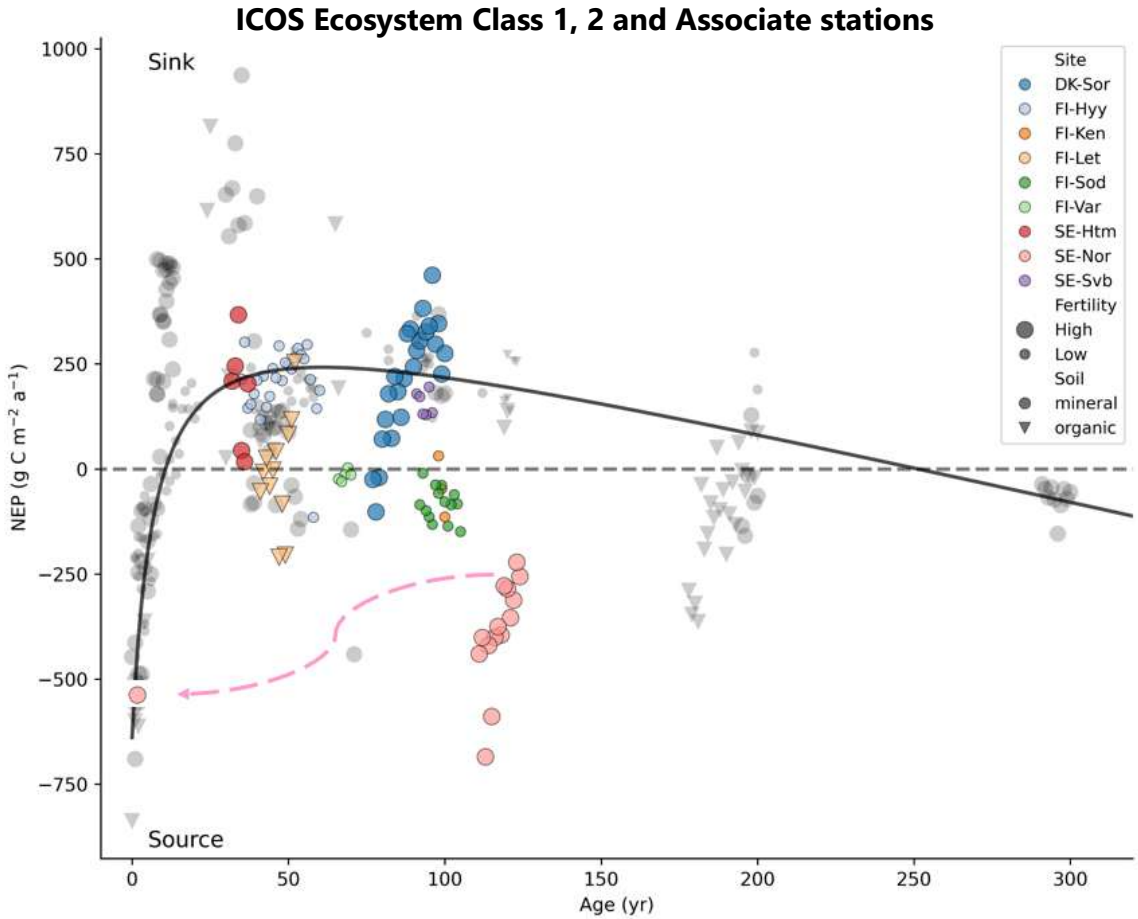
Network representativity

- ✓ ICOS Ecossystem station network == mid/late rotation coniferous forests
  - Risk of bias – C budgeting, model training, regional upscaling
- ✓ How to learn more from the few (ICOS) sites?

- ✓ We tend to neglect non-community studies & old data
- ✓ Combining 'Northern European Forest Flux' legacy data necessary



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Take home!

- Age-dependency strong in young forests
  - Source to sink ~10 yr of age, mainly due to rapid GPP recovery
  - Mid-rotation forests NEP highly site-specific
  - Old-growth forests tend to become C neutral
- GPP and ER highly correlated
  - Site fertility and temperature key drivers
  - Soil type (peat vs. mineral) non-significant in the dataset
- Large NEP on former agric. lands / pastures → afforestation benefits C balance
- Management impacts transient and recovery is fast
- NEP is highly site-specific and EC-measurements remain case-studies
- Upscaling is challenging and must be done with care

"Knowing C balance of EC-forests does not mean we know C balance of our forests in general."

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Research Council of Finland

LS-Hydro (2023-2027): From forest structure to hydrological function – merging dense EO data and process-models

PREFER (2022-2026): Precision nutrient management - a tool for mitigation of climate change and environmental loading in boreal forestry

European Commission

Horizon Europe 2021-2027

GreenFeedBack – Greenhouse gas fluxes and Earth system feedbacks

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Thank you!

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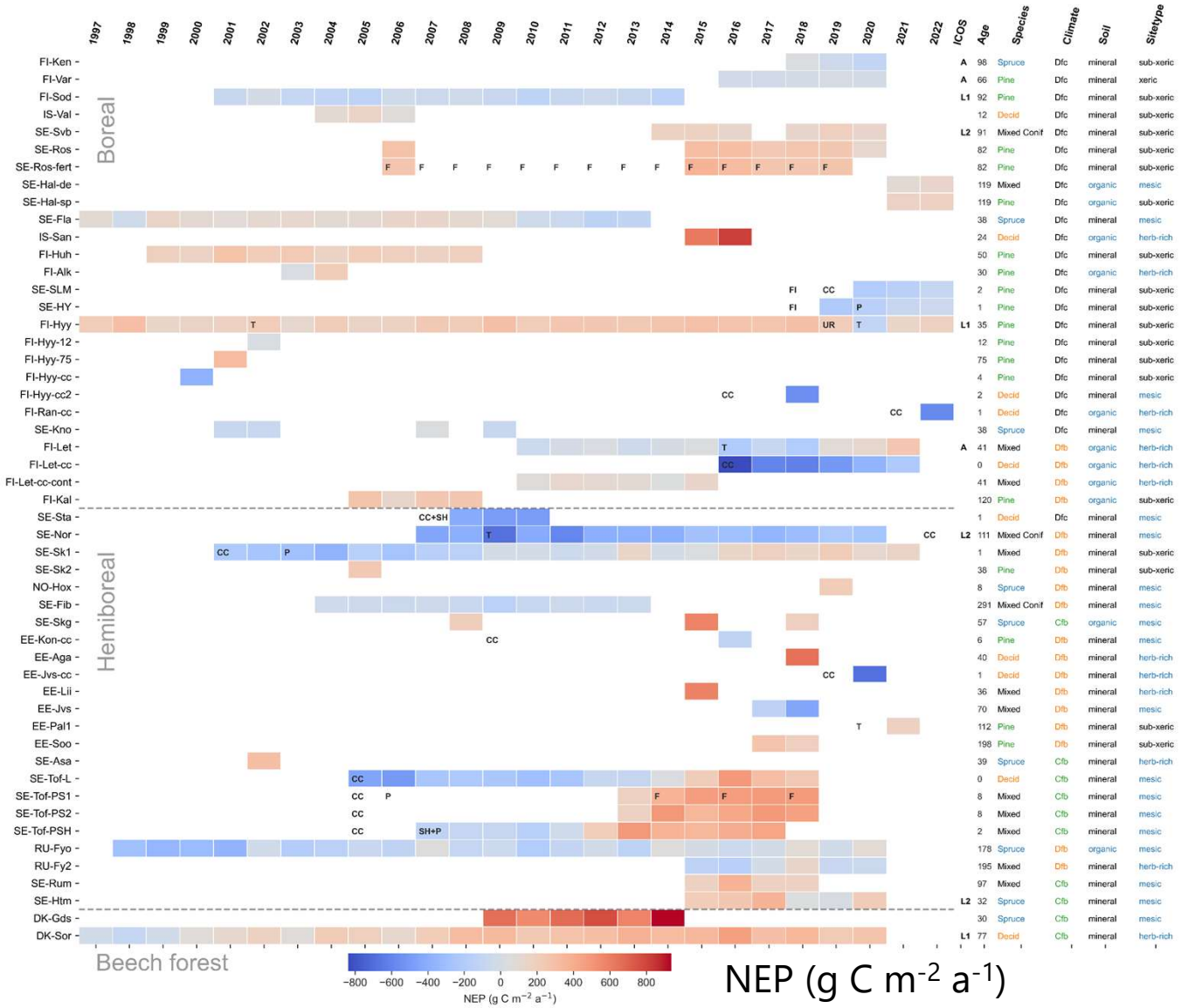

NEP (g C m^{-2} a^{-1})

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